

Code-Layout, Readability and Reusability

Date	01 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper Python indentation and HTML formatting followed.
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into templates, static, model files, and Python scripts.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive names such as application_df, credit_df, prediction, and model are used.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as predict() and Flask route methods use meaningful names.
5	Code Comments	Inline and block comments explain logic	Partial	Comments are added for important sections, but additional comments can improve readability.
6	Modular Design	Code is split into reusable functions/modules	Yes	Training (train_model.py) and application (app.py) are maintained separately.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Unnecessary and duplicate code has been removed.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic validation is present; advanced exception handling can be added in future versions.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	App.py	Python, Flask	Handles routing and prediction requests	High

2	train_model.py	Python, Scikit-learn	Model training and saving	High
3	Model.pkl	Joblib / Scikit-learn	Loaded during prediction	High
4	index.html	HTML, CSS, Jinja2	User input interface	Medium
5	result.html	HTML, CSS, Jinja2	Displays prediction results	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized folder structure with separate files for training, application, templates, and static resources.
Readability	5	Code uses meaningful variable names and follows a consistent coding style.
Reusability	5	The trained model, preprocessing files, and Flask application can be reused for future enhancements.
Documentation / Comments	4	Basic comments are included; additional documentation can further improve maintainability.
Overall Score	4.8/5	The project is well-structured, maintainable, reusable, and suitable for deployment and future development.